

CyberPatriot Ubuntu 22 Exhibition Image Answer Key

Welcome to the CyberPatriot Exhibition Round! This guide outlines how to identify and resolve common security vulnerabilities on an Ubuntu operating system to help build your cybersecurity skillset.

Note: The vulnerabilities covered in this image are fundamental. As the season progresses, competition images will become increasingly challenging. Always rely on your general cybersecurity knowledge alongside the hints provided in the image's README file.

Overview of Scored Items

Below is the breakdown of all scoring items, including detection steps, remediation instructions, and the underlying security rationale.

1. Forensics Question 1 (8 pts)

- **How to find:** Always read all Forensics Questions on the Desktop *before* modifying the system, as system changes may destroy required evidence. Open the file Forensics Question 1 on the Desktop.
- **How to solve:** The prompt asks to decode a Base64-encoded message. Use a terminal or an online base64 decoder.
Answer: affidavit
- **Why it matters:** Understanding basic encoding methods like Base64 is essential for analyzing hidden payloads or obfuscated configurations.

2. Forensics Question 2 (8 pts)

- **How to find:** Open a terminal and search for MP3 files using the locate command:
`locate '*.mp3'` The output shows non-work-related media files in Linda's Music directory.
- **How to solve:** Identify the absolute path starting from the root directory (/).
Answer: The full absolute path leading to the .mp3 files starting with the root directory /.
- **Why it matters:** Efficiently locating files via the CLI helps quickly identify unauthorized software, backdoors, or policy violations.

3. Removed Unauthorized User: jimmy (6 pts)

- **How to find:** Compare system users against the authorized list in the Desktop README file. Any account not explicitly authorized or required by system services must be removed.
- **How to solve:**
 1. Open **Settings** → **Users**.
 2. Click **Unlock** in the top right and enter your administrator password.
 3. Select user **jimmy**.
 4. Click **Remove User...** → **Delete Files**.
- **Why it matters:** Unnecessary accounts expand the system's attack surface and increase vulnerability to credential stuffing or password cracking.

4. Removed Admin Rights: doofenshmirtz (6 pts)

- **How to find:** Verify administrators against the README file. Users not listed as authorized admins must have elevated privileges revoked.
- **How to solve:**
 1. Open **Settings** → **Users**.
 2. Select user **doofenshmirtz**.
 3. Toggle off the **Administrator** switch.
- **Why it matters:** Enforcing the Principle of Least Privilege minimizes risk; compromised administrative accounts grant total host control.

5. Removed Admin Rights: vanessa (6 pts)

- **How to find:** Check the README file against current system administrators.
- **How to solve:**
 1. Open **Settings** → **Users**.
 2. Select user **vanessa**.
 3. Toggle off the **Administrator** switch.
- **Why it matters:** Restricting administrative access limits what an attacker can modify if an account gets compromised.

6. Changed Insecure Password: pinky (6 pts)

- **How to find:** Audit passwords for authorized users as instructed in the competition scenario.
- **How to solve:**
 1. Open **Settings** → **Users**.
 2. Select user **pinky**.
 3. Click **Password** → **Set a password now**.

4. Enter a strong password and click **Change**.
- **Why it matters:** Weak passwords are susceptible to offline brute-force and dictionary attacks.

7. Added User candace to Group firesidegirls (8 pts)

- **How to find:** Check special group assignments requested in the README file.
- **How to solve:** Execute the following terminal command:

```
sudo gpasswd -a candace firesidegirls
```
- **Why it matters:** Managing group permissions properly ensures users have access to necessary business tools without granting excessive privileges.

8. Enabled Uncomplicated Firewall - UFW (6 pts)

- **How to find:** Check host-based firewall status:

```
sudo ufw status
```
- **How to solve:** Enable the firewall:

```
sudo ufw enable
```
- **Why it matters:** Host-based firewalls filter unauthorized inbound and outbound network traffic.

9. Disabled Unnecessary Service: Nginx (6 pts)

- **How to find:** Check active services using:

```
systemctl list-units --type=service --state=active
```

Cross-reference active services with critical applications in the README.
- **How to solve:**

```
sudo systemctl stop nginx
```

```
sudo systemctl disable nginx
```
- **Why it matters:** Disabling unneeded background daemons reduces attack vectors and exposed network ports.

10. Configured Daily Automatic Update Checks (6 pts)

- **How to find:** Check update settings in the **Software & Updates** utility.
- **How to solve:**
 1. Open **Software & Updates**.
 2. Under the **Updates** tab, set **Automatically check for updates** to **Daily**.
- **Why it matters:** Regular update checks ensure administrators are promptly notified of critical security patches.

11. Updated Application: Firefox (5 pts)

- **How to find:** Open the **Software Updater** app to check for outdated packages.
- **How to solve:** Run **Software Updater** and install updates for Firefox.
- **Why it matters:** Web browsers frequently face public vulnerabilities that can be exploited by malicious sites.

12. Updated Application: Thunderbird (5 pts)

- **How to find:** Open the **Software Updater** app to check for outdated packages.
- **How to solve:** Run **Software Updater** and install updates for Thunderbird.
- **Why it matters:** Patching email clients prevents exploitation through malicious attachments or linked content.

13. Removed Prohibited MP3 Files (6 pts)

- **How to find:** Locate unauthorized media files:
`locate '*.mp3'`
- **How to solve:**
`sudo rm /home/linda/Music/*.mp3`
- **Why it matters:** Non-work-related media files consume storage, violate policy, and can harbor malicious code targeting media parsers.

14. Removed Prohibited Software: ophcrack (6 pts)

- **How to find:** Look for unauthorized penetration testing or cracking utilities in application menus.
- **How to solve:**
`sudo apt remove ophcrack -y`
- **Why it matters:** Password cracking tools present a major security risk when left on production machines.

15. Removed Prohibited Software: Wireshark (6 pts)

- **How to find:** Inspect installed applications for unapproved network sniffers.
- **How to solve:**
`sudo apt remove wireshark -y`
`sudo apt autoremove -y`
- **Why it matters:** Unauthorized packet analyzers allow non-admin users or attackers to sniff sensitive network credentials.

16. Secured SSH Configuration: Disabled Empty Passwords (6 pts)

- **How to find:** Inspect the SSH configuration file at `/etc/ssh/sshd_config`.
- **How to solve:**
 1. Open the configuration file:

```
sudo gedit /etc/ssh/sshd_config
```
 2. Set `PermitEmptyPasswords` no.
 3. Uncomment the line `#HostKey /etc/ssh/ssh_host_ed25519_key`.
 4. Save and exit.
- **Why it matters:** Allowing empty passwords on SSH exposes remote access endpoints to trivial authentication bypasses.

Point Penalties

Action	Penalty	Rationale
OpenSSH service stopped or removed	-5 pts	OpenSSH is designated as a critical service in the README.
Firefox removed	-5 pts	Firefox is marked as required software.
Thunderbird removed	-5 pts	Thunderbird is marked as required software.
Perl removed	-5 pts	Perl is a critical dependency and required software.